

— UC BERKELEY · ENGINEERING & ROBOTICS · CAPSTONE 2026

Navigation & Guidance of **Blind** **and Visually-Impaired** People

Camera-free swarm robotics and a cloud-connected app that give blind and visually impaired people the freedom to move independently, safely, and on their own terms — guided by two or more robots following and leading them.

Presented by **Violet** · **Samuel** · **Gabriel** · July 2026

02 · PROBLEM IDENTIFICATION

Independent indoor travel is still out of reach.

2.2B

people live with a vision impairment; blind people rise from ~43M (2020) to ~61M by 2050.

28%

of blind & partially-sighted people feel able to travel independently — even after training.

Indoors

no curbs, walls or traffic sounds as landmarks — an unfamiliar building is "too hard to attempt."

\$45–60k

a guide dog: 1–5 years to train, serves only ~2% of blind people (<19,000 working dogs, falling).

40%+

of white-cane users report head-level accidents monthly — the cane senses only below the waist.

Audio-only

BlindSquare, GoodMaps, Evelity, Wayfindr all talk — **none physically leads.**

PROBLEM STATEMENT

How might we help a blind person navigate an unfamiliar indoor space safely and independently — without a guide dog, a costly wait, or relying on another person?

03 · PROPOSED SOLUTION

Three low-cost robots that physically lead the way.

A system of three small robots plus a phone app that leads a blind traveler (Sam) along pre-mapped indoor routes. Two guide robots lead, a companion stays beside Sam, and one piece of software — the brain — coordinates all three over Bluetooth. **No camera. No internet needed.**

ASK & PLAN

- **Voice destination** — "Take me to the store."
- **Route + ETA** — "be there by 6:00 PM" and it departs on time by itself.

COORDINATE

- **Real-time (~5 Hz)** — balances speed & spacing so no one is left behind.
- **Safety** — stops on obstacles, taps the arm, speaks every step; e-stop anytime.

PRIVATE & SHARED

- **Camera-free & offline-capable** — private by design.
- **Live family view** — a relative anywhere can follow the journey.

— 04 · PROBLEM & SOLUTION STATEMENT

Stated plainly.

PROBLEM STATEMENT

- A blind or visually-impaired person **cannot travel independently** through an unfamiliar indoor space.
- Every existing aid falls short: apps are **audio-only** — they describe, they don't lead; a **guide dog** is a \$45–60k, multi-year wait serving <2%; a white cane **misses every head-height hazard**; the rest means **depending on another person**.

SOLUTION STATEMENT

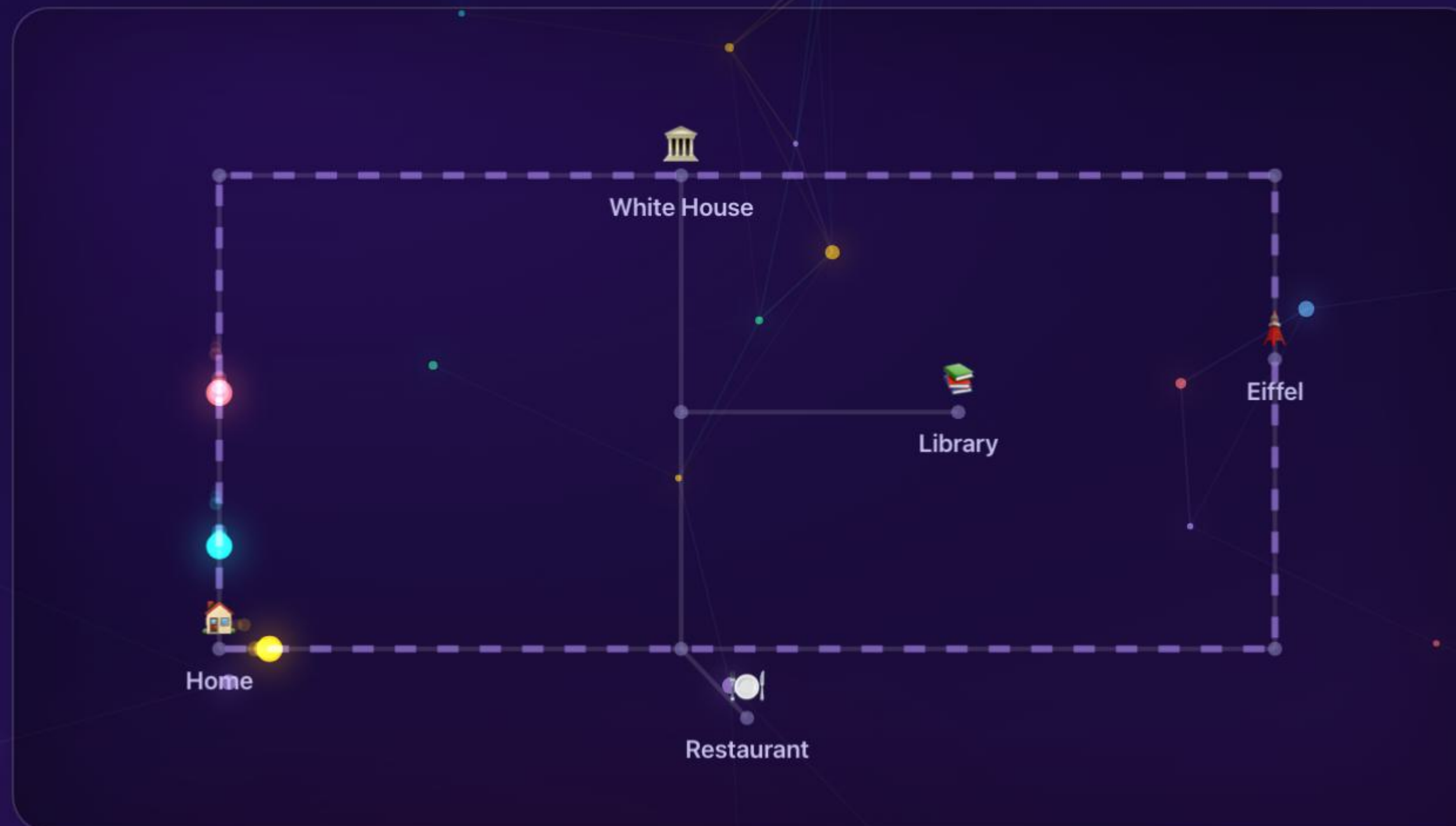
- A **camera-free swarm of three low-cost robots** that **physically lead** the traveler along a mapped indoor route.
- Two robots guide, one companion holds the group, and a **cloud-connected hub** coordinates them over Bluetooth with spoken, step-by-step guidance — **private, offline-capable, paced by the user**.

THESIS

Keep the robots simple and put the intelligence in the hub, and a coordinated swarm can deliver the one thing no audio-only aid can: **physical guidance**.

— ► THE JOURNEY — COORDINATED LIVE

Say a place. The swarm moves as one.



- Opener — leads the route
- Sam — the traveler
- Closer — holds the group

The hub plans the route, then coordinates all three **~5x per second** — balancing speed and spacing so no one is ever left behind.

What it must do — and why it matters.

OBJECTIVES

- Lead a blind user from A to B along a mapped route, **reliably**.
- Hold a safe formation — the slowest member (Sam) **sets the pace**.
- Stop at obstacles and alert the user — **light + arm tap + voice**.
- Compute ETA and support **scheduled departures**.
- Stay **low-cost, camera-free and offline-capable**.
- Provide **live monitoring** for family and caregivers.

EXPECTED IMPACT

- Restores **independent indoor mobility** for a large, ageing, under-served population.
- A kit could target **under \$2,000** — vs a \$45–60k, multi-year guide dog.
- **Physically guides** the user — unlike every audio-only app.
- Fits accessibility mandates (**EAA, ADA**) — airports, care homes, museums, hospitals.
- Assistive-independence programs return **\$4–7 per \$1 (SROI)** — plus dignity, employment, mental health.

Simple robots. One intelligent brain.



OPENER · FRONT

Scouts the route

Line + ultrasonic sensors. Leads and calls the turns.



THE TRAVELER · SAM

You, in the middle

No external sensors — the swarm does the seeing.



CLOSER · REAR

Keeps you together

Ultrasonic gap-holding so the group never breaks apart.

HARDWARE

- Each robot runs on a **Raspberry Pi Pico** (RP2350, MicroPython) — wheels + encoders, IMU, line sensors, HC-SR04 ultrasonic, servo "arm", LED.
- One identical firmware** on all three — roles assigned live.

SOFTWARE

- The hub** — a web/phone app that plans routes, coordinates the fleet, and speaks.
- Bluetooth** to the robots · **cloud sync** for the live family view.

ENGINEERED WITHIN TIGHT CONSTRAINTS

A Raspberry Pi Pico is **cheap and low-power** — little compute, little memory. So the robots stay dumb and lean, the brain lives in the hub, and the kit targets **under \$2,000** — a fraction of a \$45–60k guide dog.

Why a camera-free swarm — scored, not assumed.

CRITERION	WEIGHT	VISION ROBOT	LIDAR ROBOT	AUDIO-ONLY APP	CAMERA-FREE SWARM ▶ OURS
Privacy — no camera	20%	1	4	4	5
Affordability — <\$2k kit	20%	2	1	5	4
Physically leads the user	20%	4	4	1	5
Runs on low-cost compute	15%	1	2	5	4
Offline-capable	10%	3	4	2	5
Safety — obstacles + formation	15%	4	5	2	4
Weighted score / 5		2.45	3.25	3.25	4.50

DECISION

The swarm wins on what matters most to a blind traveler — privacy, physical guidance, offline reliability — and a Raspberry Pi Pico keeps each robot cheap. The compute limit is answered by moving the brain into the hub.

The real project is the **brain**: a hub we coded from scratch.

CLIENT

Voice interface

Phone or laptop — listens for a destination, speaks every step of the way.

CLOUD

Live state & journey logs

The hub streams state to the cloud → a remote family / caregiver view, anywhere.

THE HUB · THE BRAIN

Route planning · ~5 Hz coordination loop · reliable command protocol · voice I/O

A real, non-trivial system: a city-graph router, a swarm coordinator that balances speed & spacing, and an **acknowledged, retried, verified** command layer — the part that made it actually work.

COMMANDS ↓ · TELEMETRY ↑

SWARM · BLUETOOTH

 Opener

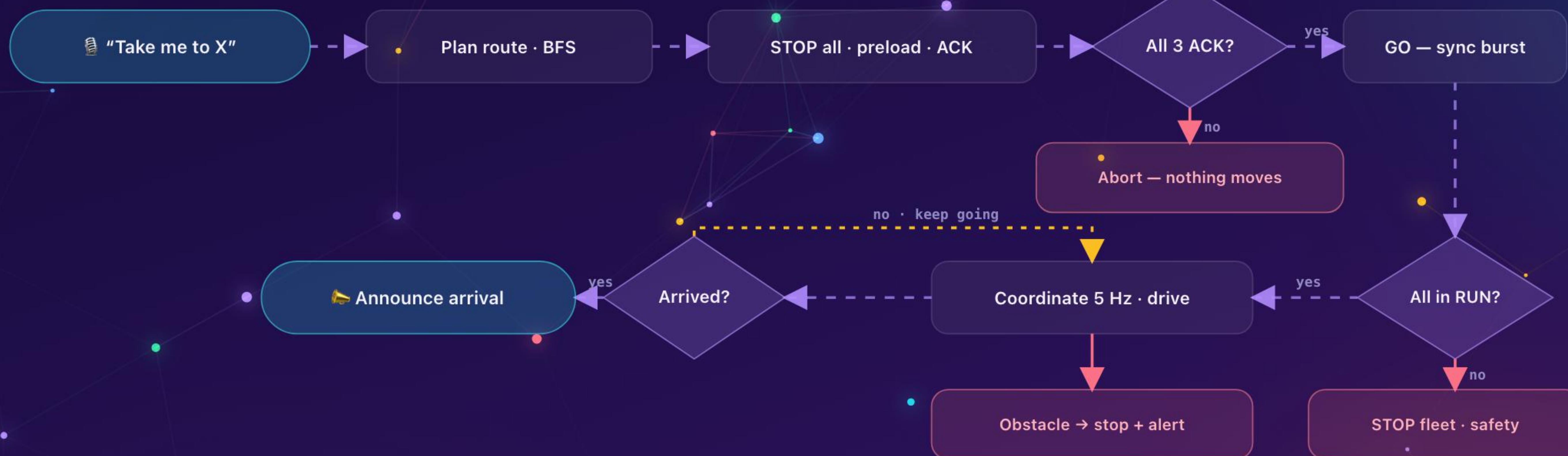
Sam

 Closer

↑ telemetry back

Open source · github.com/8b4p7b6v74-ship-it/sam-and-the-guides

How a spoken place becomes a **synchronized move**.



THE GUARANTEE

Nothing moves until all three confirm; the fleet starts in sync or safely aborts; and an obstacle stops everyone instantly.

The coordination loop, in plain steps.

```
# HUB – guide the traveler to a place
function guide(dest):
    route ← planRoute(graph, here, dest)  # BFS

    # phase 1 · preload, confirmed
    sendReliable(all, "STOP")
    for r in fleet:
        ok ← sendReliable(r, mission[r], tries=4)
        if not ok: abort("unconfirmed"); return

    # phase 2 · synchronized start
    burst(all, "GO")  # fire x3, one tick
    wait telemetry = RUN  (≤ 1.4 s)
    if not allRunning: stopAll(); return

    # phase 3 · coordinate ~5 Hz
    while not arrived:
        hold gap(Closer) → Sam.speed  # user paces
        if front(Opener) < safe: stopAll(); alert()
        on turn: mirror → Sam, then Closer
        speak(nextStep)
    speak("You have arrived.")
```

```
# ROBOT – one shared firmware
on receive(line):
    if line starts "#id":
        emit("A|" + id)  # ACK, always
        if id in seen: return  # dedup
        seen.add(id)
    cmd ← strip(line)

    if cmd = "GO":
        if mission ≠ RUN:  # idempotent
            startMission()
    elif cmd = "STOP": halt()
    elif cmd = "PATH ...": load(path)

    # telemetry, ~20 Hz
    every 50 ms:
        emit("T|role|mission|front|...")
```

WHY IT HOLDS

Every command is acknowledged & retried; GO is idempotent, so a duplicate never restarts a moving robot; and the fleet is verified in telemetry before it's trusted to move.

11 · PROTOTYPE DEVELOPMENT

From three kits to a coordinated convoy.

BUILDING PROCESS

- Assembled **3 XRP kits**; wrote **one firmware** shared by all — roles assigned live over Bluetooth.
- Built the web-based **Hub** (Web Bluetooth) that connects the fleet, plans routes on a hand-mapped model town, speaks each step, and syncs to the cloud.
- Added a **voice interface** — spoken destination in, spoken guidance out.

INITIAL TESTING

- Verified **BLE connection + role assignment** and the live telemetry loop.
- Confirmed **motor control** and per-robot state reporting.
- Built a full **software simulator** of the 3-robot convoy to validate coordination end-to-end **before touching hardware**.

The real challenge was **wireless reliability** — not the motors.

ENVIRONMENTS

- A hand-built **miniature town**, measured to the centimetre.
- A full **simulator** (virtual fleet) with **injectable packet loss**.

FINDINGS

- Commands could be **lost, delayed, or duplicated** — a robot left behind.
- Sensors were noisy — **ultrasonic spikes, encoder glitches**.

ITERATIONS

- Rebuilt comms: **acknowledgment, retry, synchronized start, verification**.
- An **adversarial review** found & fixed **10 reliability defects**.
- Hardened firmware — **idempotent commands, debounced obstacle detection**.

✓ Delivery holds at 40% simulated packet loss → still 3/3

— 13 · RESULTS & IMPACT

A swarm you can trust to move together.

SUCCESS METRICS

- Commands **confirmed by all 3** — even at 40% simulated packet loss.
- **Deterministic**: the fleet starts in sync, or safely aborts.
- Full **voice** → **coordinate** → **guide** → **arrive** validated in simulation.

USER-CENTRED DESIGN

- Camera-free & offline-capable — answers the **privacy** concern.
- The user **sets the pace** — the system adapts to them.
- Spoken guidance + arm tap for a **non-visual** experience.

CHALLENGES OVERCOME

- **BLE reliability** — a verifiable, retrying protocol.
- **Sensor noise** — filtering & debouncing.
- **Synchronization** — a two-phase, verified start.

Where it goes next.

SENSE

- **Head-height obstacle detection** — the hazard the cane misses.
- **Haptic feedback** in the companion robot.

SCALE

- **Larger swarms** and longer routes.
- **Automatic mapping** of new indoor spaces.
- Better **battery & range**.

DEPLOY

- **Real-world pilots** — airports, care homes, hospitals.
- **Trials with the people we built it for.**

Every step toward the same goal: guidance that gives independence back.

Problem identified. **Software built.** Coordination proven.

SUMMARY

- **Problem** — a large, under-served population can't move independently indoors.
- **Solution** — a camera-free, voice-first robot swarm coordinated by a cloud-connected hub.
- **Result** — a reliable, verifiable coordination layer, validated end-to-end.

LIVE DEMONSTRATION

- Speak a destination — "**Take me to the Eiffel Tower.**"
- The hub confirms all robots, then triggers a **synchronized start**.
- The fleet moves through the town and **announces arrival**.

🔗 Code & docs · github.com/8b4p7b6v74-ship-it/sam-and-the-guides

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— THANK YOU

Thank you.

To our mentors, teachers, and everyone who tested, questioned, and cheered us on — thank you. This project is for the travelers it's meant to serve.

🔗 github.com/8b4p7b6v74-ship-it/sam-and-the-guides

Violet · Samuel · Gabriel · Contact: [insert contact details]

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